

****ICSE CLASS 10 MATHEMATICS — COMPLETE REVISION NOTES****

****CHAPTER 1: GOODS AND SERVICES TAX (GST)****

****YouTube Lecture:**** https://youtu.be/iWcGaP_eWeA?si=63nqi4wBCkyehr-X

****1. Introduction to GST****

Goods and Services Tax (GST) is a comprehensive, multi-stage, destination-based tax that is levied on every value addition. It is a single indirect tax for the whole country.

****2. Key Terms in GST****

****List Price / Marked Price:**** The price of an item as quoted by the manufacturer or displayed on the price tag.

****Basic Price / Cost Price:**** The price at which a dealer buys the goods from the manufacturer, before adding any taxes.

****Input Tax Credit (ITC):**** The dealer gets credit for the tax already paid to the manufacturer on the purchase of goods.

****Output Tax:**** The tax collected by the dealer from the consumer on the sale of goods.

****GST Paid by Dealer:**** Output Tax minus Input Tax Credit.

****3. Components of GST****

GST has three components. On intra-state sales (within the same state), CGST (Central GST) and SGST (State GST) are levied, equally splitting the total GST rate. On inter-state sales (between different states), IGST (Integrated GST) is levied.

For example, if the GST rate is 18% on an intra-state sale, CGST = 9% and SGST = 9%.

****4. Computation of GST****

(a) The manufacturer sells goods to a dealer. GST is added to the manufacturer's price. The dealer pays this total amount.

(b) The dealer sells goods to a consumer. GST is added to the dealer's selling price.

(c) The dealer deposits the net GST: Total Output Tax (collected from consumer) minus Input Tax Credit (already paid to the manufacturer).

(d) The total amount paid by the consumer is the final price including GST. This is the sum of the dealer's price and the total applicable GST.

****5. Problems Involving Discounts, Profit and Loss****

Generally, the discount is first applied to the List Price / Marked Price to get the taxable value. GST is then calculated on this discounted selling price. Profit or loss is computed on the cost price, and the GST calculation is on the selling price.

****IMPORTANT QUESTIONS: GOODS AND SERVICES TAX (GST)****

****Q1.**** A manufacturer sells a washing machine to a dealer for Rs 15,000. The dealer sells it to a consumer at a profit of Rs 3,000. If the rate of GST is 12%, calculate:

- (a) The amount of tax (under CGST and SGST) paid by the dealer to the manufacturer.
- (b) The amount of tax (under CGST and SGST) collected by the dealer from the consumer.
- (c) The net tax paid by the dealer to the government.
- (d) The final price paid by the consumer.

****Q2.**** The list price of a television is Rs 40,000. The shopkeeper sells it to a consumer at a discount of 10%. If the rate of GST is 18%, find:

- (a) The selling price after discount.
- (b) The amount of CGST and SGST.
- (c) The final amount paid by the consumer.

****Q3.**** A dealer in Ahmedabad (Gujarat) sells an article to a consumer in Bhopal (Madhya Pradesh) for Rs 8,000. If the rate of IGST is 18%, calculate:

- (a) The amount of IGST.
- (b) The final price paid by the consumer.

****Q4.**** A shopkeeper buys an article whose list price is Rs 2,000 at a discount of 20% on the list price from a wholesaler. The shopkeeper sells the article to a consumer at the list price. If the rate of GST is 12%, calculate the net GST paid by the shopkeeper to the government.

****Q5.**** For a dealer, the cost price of an article is Rs X. He marks it up by some amount, sells it and pays net GST. If the GST rate and final price are given, be able to compute the input and output tax and the profit.

****Q6.**** A manufacturer sells a pair of shoes costing Rs 1,500 to a distributor at a profit of 20%. The distributor sells it to a retailer at a profit of 10%. The retailer sells it to a consumer at a profit of 15%. Find the total GST liability of each person and the final price paid by the consumer if the GST rate is 12%.

****CHAPTER 2: BANKING****

****YouTube Lecture:**** <https://youtu.be/gZCrI573Yy8?si=9y3mhkgisFuYM7g2>

****1. Recurring Deposit Account****

A Recurring Deposit (RD) account is an account in which a fixed sum of money is deposited every month for a specified period. After the maturity period, the total accumulated amount, including interest, is returned to the depositor.

****2. Key Terms****

****P:**** Monthly installment (deposit).

****n:**** Number of months for which the deposit is made.

****r:**** Rate of interest per annum.

****I:**** Total interest earned.

****MV:**** Maturity Value.

****3. Formulae for Recurring Deposit****

The total principal for the purpose of interest calculation is considered as if it was deposited for the equivalent period. Using the standard formula:

$$I = P \times [n(n+1) / (2 \times 12)] \times (r / 100)$$

$$\text{Simplified: } I = P \times [n(n+1) / 24] \times (r / 100)$$

$$MV = (P \times n) + I$$

This formula works because the first deposit earns interest for n months, the second for (n-1) months, and so on, which when summed gives the equivalent of the principal P deposited for $n(n+1)/2$ months.

****IMPORTANT QUESTIONS: BANKING****

****Q1.**** Nagma has a Recurring Deposit account in a bank for 3 years at 8% p.a. If she gets Rs 9,990 as interest, find her monthly installment.

****Q2.**** Mr. Saluja deposits Rs 800 per month in a Recurring Deposit account for a period of 2 years. At the time of maturity, he receives Rs 20,600. Find the rate of interest paid by the bank.

****Q3.**** A man opened a recurring deposit account in a bank and deposited Rs 500 per month for 2 years. If he received Rs 15,000 at the time of maturity, find the rate of interest per annum.

****Q4.**** Shobha has a recurring deposit account and deposits Rs 600 per month for 4 years. If the rate of interest is 10% p.a., find the amount she will receive at the time of maturity.

****Q5.**** A person deposits a certain sum every month in a bank for n months. The rate of interest is r% p.a. and the maturity value is V. Set up an equation to find the monthly deposit P.

****Q6.**** Suraj deposits Rs 1,000 every month in a recurring deposit account for 3 years. The bank pays interest at 8% p.a. Find the total amount due to Suraj at maturity.

****CHAPTER 3: SHARES AND DIVIDENDS****

****YouTube Lecture:**** <https://youtu.be/cxuL191S6VY?si=Y-sxqIO85YTTIXZk>

****1. Introduction to Shares****

A company raises capital by issuing shares to the public. A person who buys these shares is a shareholder and becomes a part owner of the company.

****2. Key Terms****

****Face Value (FV) / Nominal Value:**** The original value of a share as stated by the company. This is fixed.

****Market Value (MV):**** The price at which a share is bought or sold in the stock market. It fluctuates based on demand and supply.

****At Par:**** When $MV = FV$.

****At Premium:**** When $MV > FV$. $Premium = MV - FV$.

****At Discount:**** When $MV < FV$. $Discount = FV - MV$.

****Dividend:**** The part of the company's profit distributed to shareholders. It is always declared as a percentage of the Face Value.

****Rate of Dividend:**** The percentage of face value paid as dividend.

****3. Important Formulae****

$$\text{Number of Shares} = \text{Total Investment} / \text{Market Value per share}$$

$$\text{Annual Income from shares} = (\text{Number of Shares}) \times (\text{Rate of Dividend} / 100) \times (\text{Face Value per share})$$

$$\text{Total Investment} = \text{Number of Shares} \times \text{Market Value per share}.$$

$$\text{Rate of Return on Investment (\%)} = (\text{Annual Income} / \text{Total Investment}) \times 100\%$$

****IMPORTANT QUESTIONS: SHARES AND DIVIDENDS****

****Q1.**** A man invests Rs 8,800 in shares of a company which pays 10% dividend. A share has a face value of Rs 100 and is available at a market price of Rs 110.

(a) Find the number of shares bought.

(b) Find the annual income from the shares.

(c) Find the rate of return on his investment.

****Q2.**** A person wants to earn an annual income of Rs 1,500 by investing in shares of face value Rs 100 each available at Rs 125 each. If the company pays a dividend of 12%, how much money should he invest?

****Q3.**** Which is a better investment:
12% at Rs 120 (FV Rs 100), or
8% at Rs 90 (FV Rs 100)?

****Q4.**** Mr. Shukla has 200 shares of a company. The company declares a 15% dividend. If the face value of each share is Rs 10, find his total dividend income.

****Q5.**** A company declares a dividend of 8%. A man receives Rs 1,200 as his annual dividend. If the face value of each share is Rs 10, find the number of shares he holds.

****Q6.**** A man sells 500 shares (FV Rs 10) at Rs 45 and invests the proceeds in 15% shares of Rs 100 at Rs 125. Find the change in his annual income.

****CHAPTER 4: LINEAR INEQUALITIES****

****YouTube Lecture:**** https://youtu.be/gzwo0IRLkM4?si=was8nYqWJFh_tHCr

****1. Introduction to Linear Inequalities****

A linear inequality is a mathematical statement expressing inequality between two algebraic expressions. It uses signs like:

- < (less than)
- > (greater than)
- ≤ (less than or equal to)
- ≥ (greater than or equal to)

****2. Rules for Solving Inequalities****

The same number can be added or subtracted from both sides without changing the inequality.

Both sides can be multiplied or divided by the same positive number without changing the inequality.

When both sides are multiplied or divided by a negative number, the sign of the inequality reverses. Example: $-x > 3$ becomes $x < -3$.

****3. Replacement Set and Solution Set****

The replacement set is the set of numbers from which the solution values are to be chosen. It is usually specified as N (Natural Numbers), W (Whole Numbers), Z (Integers), or R (Real

Numbers). The solution set is the set of all values from the replacement set that satisfy the inequation.

****4. Representation of Solution on the Number Line****

The solution set is shown on a number line.

A filled dot (solid circle) represents that the end value is included ($\leq$ or $\geq$).

An open dot or hollow circle represents that the end value is excluded ($<$ or $>$).

****IMPORTANT QUESTIONS: LINEAR INEQUATIONS****

****Q1.**** Solve the following inequation and represent the solution set on the number line:

$2x - 3 < 5$, x belongs to N (Natural Numbers)

****Q2.**** Solve: $-3x + 4 \geq -5$, x belongs to Z . Represent on number line.

****Q3.**** Solve the inequation and graph the solution on the number line:

$3x - 1 \geq 5$, x belongs to W

****Q4.**** Find the solution set of the inequality $5 - 2x \leq x - 4$, given that x is a whole number.

****Q5.**** Solve for x : $-2 < (x/3) \leq 2$, and find the solution when x is an Integer and when x is a Real number.

****Q6.**** Represent the following on a number line:

The set of numbers x : $-3 \leq x < 4$, x belongs to R .

****Q7.**** For an inequation $3 \leq 2x - 1 < 7$, find the range of x if x belongs to R . Also list the integers within this range that satisfy the inequality.

****Q8.**** Solve for x when the replacement set is Natural Numbers: $4x + 3 > 2x + 9$.

****CHAPTER 5: QUADRATIC EQUATIONS****

****YouTube Lecture:**** <https://youtu.be/q7TmhgBbEJ8?si=JFJLWbjTDX7XXL4b>

****1. Standard Form of a Quadratic Equation****

A quadratic equation in variable x is an equation of the form:

$$ax^2 + bx + c = 0$$

where a , b , c are real numbers and a is not equal to 0.

****2. Nature of Roots (Using Discriminant)****

The discriminant (D) is given by:

$$D = b^2 - 4ac$$

If $D > 0$, the equation has two distinct real roots.

If $D = 0$, the equation has two equal real roots (one real root).

If $D < 0$, the equation has no real roots (roots are imaginary).

****3. Methods of Solving Quadratic Equations****

****A. By Factorisation****

Split the middle term into two parts such that their sum is b and their product is ac . Factor by grouping and set each factor to zero to find the roots.

****B. By Using the Quadratic Formula****

The roots of $ax^2 + bx + c = 0$ are given by:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

****4. Solving Quadratic Equation Problems****

Many word problems can be translated into quadratic equations. Translate the given condition into an equation, solve it using the formula or factorisation, and reject any extraneous root (e.g., negative age, negative length).

****IMPORTANT QUESTIONS: QUADRATIC EQUATIONS****

****Q1.**** Without solving, find the nature of roots of the quadratic equation $2x^2 - 7x + 3 = 0$.

****Q2.**** Determine the value of k for which the equation $4x^2 - 8x + k = 0$ has equal roots.

****Q3.**** Solve the quadratic equation $x^2 - 5x + 6 = 0$ by factorisation.

****Q4.**** Using the quadratic formula, solve $3x^2 + 5x - 2 = 0$.

****Q5.**** The sum of a number and its reciprocal is $17/4$. Find the number.

****Q6.**** The speed of a boat in still water is 15 km/h. It goes 30 km upstream and returns downstream in 4 hours 30 minutes. Find the speed of the stream.

****Q7.**** A two digit number is such that the product of its digits is 18. When 63 is subtracted from the number, the digits interchange their places. Find the number.

****Q8.**** Solve for x : $\frac{(x-3)}{(x+3)} - \frac{(x+3)}{(x-3)} = \frac{48}{7}$, x not equal to 3, -3.

****CHAPTER 6: RATIO AND PROPORTION****

****YouTube Lecture:**** https://youtu.be/9yFiifEYiPI?si=WPxEG3iset6_Lbw1

****1. Basic Concepts****

****Ratio:**** Comparison of two quantities of the same kind. Written as $a:b$ or a/b .

****Proportion:**** Equality of two ratios. If $a:b = c:d$, then a, b, c, d are in proportion.

****Continued Proportion:**** Three quantities a, b, c are in continued proportion if $a:b = b:c$.

Here, b is called the mean proportion between a and c .

$$b^2 = ac$$

****2. Properties of Proportion****

For a proportion $a/b = c/d$:

****Invertendo:**** $b/a = d/c$

****Alternendo:**** $a/c = b/d$

****Componendo:**** $(a+b)/b = (c+d)/d$

****Dividendo:**** $(a-b)/b = (c-d)/d$

****Componendo and Dividendo:**** $(a+b)/(a-b) = (c+d)/(c-d)$

****IMPORTANT QUESTIONS: RATIO AND PROPORTION****

****Q1.**** If $(3a + 2b) : (5a + 3b) = 18 : 29$, find $a : b$.

****Q2.**** Find the mean proportional between 4 and 25.

****Q3.**** If x, y, z are in continued proportion, prove that $(x + y)^2 / (y + z)^2 = x / z$.

****Q4.**** Using the properties of proportion, solve for x :

$$(x^2 + 1) / (x^2 - 1) = 5 / 4$$

****Q5.**** If $(7x - 15y) : (3x + 9y) = 11 : 7$, find $x : y$.

****Q6.**** What number must be added to each of the numbers 6, 15, 20 and 43 to make them proportional?

****Q7.**** If $a/b = c/d$, prove that $(a^2 + c^2) / (b^2 + d^2) = ac / bd$.

****CHAPTER 7: REMAINDER AND FACTOR THEOREM****

****YouTube Lecture:**** <https://youtu.be/KUmUXT0TIY8?si=2YkVvAzzVbbbxYeA>

****1. Remainder Theorem****

If a polynomial $f(x)$ is divided by $(x - a)$, the remainder R is equal to $f(a)$.

For example, if $f(x)$ is divided by $(x + 2)$, put $x = -2$ in $f(x)$ to find the remainder.

****2. Factor Theorem****

If a polynomial $f(x)$ leaves remainder 0 when divided by $(x - a)$, i.e., if $f(a) = 0$, then $(x - a)$ is a factor of $f(x)$. Conversely, if $(x - a)$ is a factor of $f(x)$, then $f(a) = 0$.

****3. Factorising a Polynomial Completely****

Use the factor theorem to find one factor (usually by trial and error on simple values like $x = 1, -1, 2, -2$). Once one factor is found, divide the polynomial by that factor (long division or synthetic division) to reduce it to a quadratic factor. Then factorise the quadratic factor completely.

****IMPORTANT QUESTIONS: REMAINDER AND FACTOR THEOREM****

****Q1.**** Use the Remainder Theorem to find the remainder when $x^3 - 2x^2 + 4x - 8$ is divided by $(x - 2)$.

****Q2.**** Determine, using the Factor Theorem, whether $(x + 1)$ is a factor of $2x^3 + 5x^2 - x - 6$.

****Q3.**** If $(x - 3)$ is a factor of $x^3 - kx^2 + 3x + 9$, find the value of k .

****Q4.**** Factorise completely: $x^3 - 3x^2 - x + 3$.

****Q5.**** Show that $(x - 1)$, $(x + 2)$ and $(x - 3)$ are factors of $x^3 - 2x^2 - 5x + 6$ and factorise it completely.

****Q6.**** Find the values of a and b if $(x + 1)$ and $(x - 2)$ are factors of $x^3 + ax^2 + bx + 6$.

****CHAPTER 8: MATRICES****

****YouTube Lecture:**** <https://youtu.be/JwLBFvw5owU?si=7lg7pznk1Ua6Y5S>

****1. Introduction to Matrices****

A matrix is a rectangular arrangement of numbers (entries) in rows and columns. The order of a matrix is $m \times n$, where m is the number of rows and n is the number of columns.

****2. Types of Matrices****

****Row Matrix:**** Only one row ($1 \times n$).

****Column Matrix:**** Only one column ($m \times 1$).

****Square Matrix:**** Number of rows equals number of columns ($m = n$, like 2×2).

****Null (Zero) Matrix:**** All elements are zero.

****Identity Matrix (I):**** A square matrix with 1s on the main diagonal and 0s elsewhere.

For 2×2 Identity Matrix I:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

****3. Matrix Operations****

****Addition and Subtraction:**** Can be done only if the matrices are of the same order. Corresponding elements are added or subtracted.

****Scalar Multiplication:**** Each element of the matrix is multiplied by the scalar (number).

****Matrix Multiplication:**** Two matrices A and B can be multiplied only if the number of columns of A equals the number of rows of B.

****4. Multiplication of 2×2 Matrices****

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} p & q \\ r & s \end{bmatrix}$, then AB is computed by multiplying rows of A with columns of B.

$$AB = \begin{bmatrix} ap+br & aq+bs \\ cp+dr & cq+ds \end{bmatrix}$$

****IMPORTANT QUESTIONS: MATRICES****

****Q1.**** Given matrix $A = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 0 \\ 2 & -2 \end{bmatrix}$, find $A + B$ and $3A - 2B$.

****Q2.**** Find the product AB where $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix}$.

****Q3.**** If $\begin{bmatrix} a & 2b \\ c & 3d \end{bmatrix} + \begin{bmatrix} -1 & 4 \\ 5 & -2 \end{bmatrix} = \begin{bmatrix} 2 & 6 \\ 7 & 1 \end{bmatrix}$, find a , b , c , and d .

****Q4.**** Given $A = \begin{bmatrix} 4 & 2 \\ 1 & 0 \end{bmatrix}$, find $A^2 - 4A$.

****Q5.**** Solve for X if $2X + \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 5 & 4 \end{bmatrix}$.

****Q6.**** Show that matrix $A = \begin{bmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{bmatrix}$ satisfies $A \times A^T = I$.

****CHAPTER 9: ARITHMETIC PROGRESSION (AP)****

****YouTube Lecture:**** <https://youtu.be/1HxhsNvpzc?si=i0s8zBgJWkjYvrMu>

****1. Definition of Arithmetic Progression****

An Arithmetic Progression (AP) is a sequence of numbers in which the difference between any two consecutive terms is constant. This constant difference is called the common difference (d).

****2. Key Terms and Formulas****

****First Term:**** a

****Common Difference:**** d

****nth Term (General Term):****

$$T_n = a + (n - 1)d$$

****Sum of first n Terms (S_n):****

$$S_n = (n/2) [2a + (n - 1)d]$$

Also, $S_n = (n/2) [a + l]$, where l is the last term.

****3. Simple Applications****

Find the nth term when a, d, n are given.

Find the sum of n terms when a, d, n are given.

Find the common difference if two terms are given.

Find n if sum, first term and last term are given.

****IMPORTANT QUESTIONS: ARITHMETIC PROGRESSION (AP)****

****Q1.**** Find the 20th term of the AP: 5, 9, 13, 17, ...

****Q2.**** Which term of the progression 20, 18, 16, ... is -80?

****Q3.**** Determine the AP whose third term is 16 and the 7th term exceeds the 5th term by 12.

****Q4.**** Find the sum of the first 25 terms of the AP: -7, -4, -1, 2, ...

****Q5.**** How many terms of the AP 9, 17, 25, ... must be taken to make a sum of 636?

****Q6.**** The first term of an AP is 5, the last term is 45, and the sum of all terms is 400. Find the number of terms and the common difference.

****Q7.**** Find the sum of all natural numbers between 100 and 500 which are divisible by 7.

****CHAPTER 10: GEOMETRIC PROGRESSION (GP)****

****YouTube Lecture:**** https://youtu.be/QR9B3qMXDDY?si=HBsz7GT3k1Q_iwHG

****1. Definition of Geometric Progression****

A Geometric Progression (GP) is a sequence of numbers in which the ratio of any term to its preceding term is constant. This constant ratio is called the common ratio (r).

Condition: $r = T_2/T_1 = T_3/T_2 = \dots$

****2. Key Terms and Formulas****

****First Term:**** a

****Common Ratio:**** r

****nth Term (General Term):****

$$T_n = ar^{(n-1)}$$

****Sum of first n Terms (S_n):****

$$\text{If } r > 1: S_n = a(r^n - 1) / (r - 1)$$

$$\text{If } r < 1: S_n = a(1 - r^n) / (1 - r)$$

****IMPORTANT QUESTIONS: GEOMETRIC PROGRESSION (GP)****

****Q1.**** Find the 7th term of the GP: 3, 6, 12, 24, ...

****Q2.**** For a GP, the first term is 5 and the common ratio is 2. Find its 10th term.

****Q3.**** Determine the number of terms in the GP: 2, 6, 18, 54, ..., 4374.

****Q4.**** Find the sum of the first 8 terms of the GP: 2, 6, 18, ...

****Q5.**** The third term of a GP is 24 and the sixth term is 192. Find the 10th term.

****Q6.**** Find the sum of the series: $1 + 3 + 9 + 27 + \dots$ to 10 terms.

****Q7.**** In a GP, the sum of the first three terms is 21 and their product is 216. Find the terms.

****CHAPTER 11: REFLECTION****

****YouTube Lecture:**** https://youtu.be/N51MhaYu18I?si=_WFF-1ucMySzs-pL

****1. Reflection in Coordinate Geometry****

Reflection is the mirror image of a point or a shape across a given line. The size and shape of the object remain unchanged after reflection.

****2. Reflection of a Point in Different Lines****

****In the x-axis ($y = 0$):**** (x, y) becomes $(x, -y)$

****In the y-axis ($x = 0$):**** (x, y) becomes $(-x, y)$

****In the origin $(0, 0)$:** (x, y) becomes $(-x, -y)$

****In a vertical line $x = a$:** (x, y) becomes $(2a - x, y)$

****In a horizontal line $y = a$:** (x, y) becomes $(x, 2a - y)$

****3. Invariant Points****

A point is invariant under reflection if it does not change its position after reflection.

Under reflection in the x-axis: Points on the x-axis, i.e., $(k, 0)$, are invariant.

Under reflection in the y-axis: Points on the y-axis, i.e., $(0, k)$, are invariant.

Under reflection in the origin: Only the point $(0, 0)$ is invariant.

****IMPORTANT QUESTIONS: REFLECTION****

****Q1.**** Find the coordinates of the image of the point $P(5, -3)$ under reflection in the x-axis and in the y-axis.

****Q2.**** A point $P(a, b)$ is reflected in the x-axis to $P'(5, -2)$. Find the values of a and b .

****Q3.**** Plot the points $A(1, 4)$, $B(3, 4)$, $C(3, 1)$ and $D(1, 1)$ on a graph. Reflect the figure in the y-axis and write the coordinates of the image points.

****Q4.**** A point (p, q) is invariant under reflection in the line $x = 3$. Find the value of p .

****Q5.**** Reflect the point $(-4, 3)$ in the origin. Then reflect the image in the line $y = 1$. Write the final coordinates.

****Q6.**** Point $P(-2, b)$ is reflected as $P'(a, -4)$ under reflection in the origin. Find a and b .

****CHAPTER 12: SECTION AND MID-POINT FORMULA****

****YouTube Lecture:**** <https://youtu.be/f5IVhHnfyCk?si=MJYZUIAv0DH8-J7U>

****1. Distance Formula****

The distance between two points P(x1, y1) and Q(x2, y2) is:

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

****2. Section Formula (Internal Division)****

If a point P(x, y) divides the line segment joining A(x1, y1) and B(x2, y2) internally in the ratio m : n, then:

$$x = \frac{m x_2 + n x_1}{m + n}$$

$$y = \frac{m y_2 + n y_1}{m + n}$$

****3. Mid-point Formula****

The coordinates of the midpoint M of the line segment joining A(x1, y1) and B(x2, y2) are:

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

****4. Centroid of a Triangle****

The centroid G of a triangle with vertices A(x1, y1), B(x2, y2), C(x3, y3) is given by:

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

****IMPORTANT QUESTIONS: SECTION AND MID-POINT FORMULA****

****Q1.**** Find the distance between the points (2, 3) and (6, 6).

****Q2.**** Find the ratio in which the point P(-1, 6) divides the line segment joining A(-3, 10) and B(6, -8).

****Q3.**** If the midpoint of the line segment joining A(8, -6) and B(x, y) is (4, -2), find the coordinates of B.

****Q4.**** The coordinates of one end of a diameter of a circle are (4, -1) and the centre of the circle is (1, -3). Find the coordinates of the other end of the diameter.

****Q5.**** Find the centroid of the triangle whose vertices are A(2, 3), B(-1, -2), C(5, 1).

****Q6.**** A(2, 5), B(-1, 2) and C(5, 8) are the vertices of a triangle. Find the length of the median through A.

****CHAPTER 13: EQUATION OF A STRAIGHT LINE****

****YouTube Lecture:**** <https://youtu.be/UXYgRZWDgII?si=KRYQQnwxQDUEX4y8>

****1. Slope (Gradient) of a Line****

The slope 'm' of a line is the tangent of the angle it makes with the positive direction of the x-axis, measured anticlockwise.

$$m = \tan \theta$$

For two points (x_1, y_1) and (x_2, y_2) :

$$m = (y_2 - y_1) / (x_2 - x_1)$$

****2. Forms of Equation of a Line****

****Slope-Intercept Form:****

$$y = mx + c$$

where 'c' is the y-intercept (point where line cuts y-axis, $x = 0$).

****Point-Slope Form:****

$$y - y_1 = m(x - x_1)$$

****Two-Point Form:****

$$y - y_1 = [(y_2 - y_1)/(x_2 - x_1)] (x - x_1)$$

****3. Conditions for Parallel and Perpendicular Lines****

****Parallel Lines:**** Slopes are equal.

$$m_1 = m_2$$

****Perpendicular Lines:**** Product of slopes is -1.

$$m_1 \times m_2 = -1$$

****IMPORTANT QUESTIONS: EQUATION OF A STRAIGHT LINE****

****Q1.**** Find the slope and y-intercept of the line $3x - 4y = 12$.

****Q2.**** Find the equation of a line passing through the point $(-2, 3)$ with slope 4.

****Q3.**** Find the equation of the line joining the points $(2, 3)$ and $(4, 7)$.

****Q4.**** Show that the points $(1, -2)$, $(3, 0)$, $(1, 2)$ and $(-1, 0)$ form a square or find the equations of its diagonals.

****Q5.**** Find the equation of a line parallel to $2x + 3y = 7$ and passing through $(-1, 4)$.

****Q6.**** The line through A(2, 3) and B(4, x) is perpendicular to the line $3x - y + 5 = 0$. Find x.

****Q7.**** Find the equation of the line which makes an angle of 30 degrees with the positive x-axis and has a y-intercept of -2.

****CHAPTER 14: SIMILARITY****

****YouTube Lecture:**** <https://youtu.be/kfxibvVquHw?si=kGmBZFvJPtYEqRGe>

****1. Similarity of Triangles****

Two triangles are similar if their corresponding sides are proportional (same shape, not necessarily the same size).

****2. Conditions for Similarity****

****SAS (Side Angle Side):**** One angle of one triangle equals one angle of the other, and the sides containing these angles are proportional.

****SSS (Side Side Side):**** All corresponding sides are in the same ratio.

****AA (Angle Angle):**** Two angles of one triangle are equal to two angles of the other.

****3. Basic Proportionality Theorem (BPT)****

If a line is drawn parallel to one side of a triangle intersecting the other two sides, it divides those sides in the same ratio.

For triangle ABC, if DE is parallel to BC, then:

$$AD/DB = AE/EC$$

****4. Areas of Similar Triangles****

The ratio of the areas of two similar triangles is equal to the ratio of the squares of their corresponding sides.

$$\text{Area(Triangle 1)} / \text{Area(Triangle 2)} = (\text{Side 1} / \text{Side 2})^2$$

****IMPORTANT QUESTIONS: SIMILARITY****

****Q1.**** In triangle ABC, D and E are points on AB and AC respectively such that DE is parallel to BC. If AD = 2 cm, DB = 3 cm, AE = (3x - 2) cm and EC = (5x - 4) cm, find x.

****Q2.**** Prove that the areas of two similar triangles are proportional to the squares of their corresponding sides.

****Q3.**** In two similar triangles PQR and LMN, $QR = 15$ cm and $MN = 18$ cm. If the area of PQR is 225 cm^2 , find the area of LMN.

****Q4.**** A vertical pole of height 6 m casts a shadow 4 m long on the ground. At the same time, a tower casts a shadow 28 m long. Find the height of the tower (using similarity).

****Q5.**** Prove that the ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding medians.

****Q6.**** In triangle ABC, DE is parallel to BC, with D on AB and E on AC. $AD = 1.5$ cm, $DB = 3$ cm, $AE = 1$ cm. Find EC.

****CHAPTER 15: CIRCLES – CYCLIC PROPERTIES****

****YouTube Lecture:**** <https://youtu.be/No8okxgy0vA?si=8mJXeQY9Q94EcR4v>

****1. Cyclic Quadrilateral****

A quadrilateral with all its four vertices lying on a circle is called a cyclic quadrilateral.

****2. Key Cyclic Properties****

****Opposite Angles:**** The sum of opposite angles of a cyclic quadrilateral is 180 degrees (Supplementary).

$$\text{Angle A} + \text{Angle C} = 180 \text{ degrees}$$

$$\text{Angle B} + \text{Angle D} = 180 \text{ degrees}$$

****Exterior Angle:**** The exterior angle at a vertex of a cyclic quadrilateral is equal to the interior opposite angle. If AB is produced to E, then:

$$\text{Angle CBE} = \text{Angle ADC}$$

****3. Using Cyclic Properties****

To prove that four points are concyclic (lie on the same circle), show that the angles subtended by a segment at two points on the same side of it are equal, or that opposite angles of the quadrilateral formed by them sum to 180 degrees.

****IMPORTANT QUESTIONS: CIRCLES – CYCLIC PROPERTIES****

****Q1.**** In a cyclic quadrilateral ABCD, Angle A = $(x + 10)$ degrees, Angle B = $(y + 20)$ degrees, Angle C = $(3y - 30)$ degrees and Angle D = $(2x - 20)$ degrees. Find the values of x and y and hence all four angles.

****Q2.**** In the given figure, ABCD is a cyclic quadrilateral. Side AB is produced to E. If Angle CBE = 110 degrees and Angle ADC = 85 degrees, find Angle DAB and Angle DCB.

****Q3.**** Prove that the sum of opposite angles of a cyclic quadrilateral is 180 degrees.

****Q4.**** Two chords AB and CD of a circle intersect at point P outside the circle. Prove that triangles APC and BPD are similar. This question uses the external angle property.

****Q5.**** ABCD is a cyclic trapezium with AB parallel to DC. Show that AD = BC.

****Q6.**** In a circle, chords AC and BD intersect at E. If Angle AED = 120 degrees and Angle BDC = 40 degrees, find Angle BAC.

****CHAPTER 16: CIRCLES – ANGLE PROPERTIES****

****YouTube Lecture:**** <https://youtu.be/nSuSvnCEGZg?si=qDMhnkHeBpoXNqk0>

****1. Angle Subtended by an Arc at the Centre****

The angle which an arc of a circle subtends at the centre is double the angle it subtends at any point on the remaining part of the circle.

For arc AB: Angle AOB (at centre) = 2 x Angle ACB (at any point on the circle)

****2. Angles in the Same Segment****

Angles subtended by an arc in the same segment of a circle are equal. All angles formed on the same side of a chord and in the same arc of the circle are equal to each other.

****3. Angle in a Semi-Circle****

The angle in a semi-circle is a right angle (90 degrees). If AB is the diameter, then Angle ACB = 90 degrees for any point C on the circle.

****IMPORTANT QUESTIONS: CIRCLES – ANGLE PROPERTIES****

****Q1.**** Prove that the angle subtended by an arc at the centre is double the angle subtended by it at any point on the remaining part of the circle.

****Q2.**** In a circle of radius 5 cm, AB is a chord. The angle subtended by chord AB at the centre is 60 degrees. Find the length of chord AB and its distance from the centre.

****Q3.**** In the given figure, O is the centre of the circle and Angle AOC = 120 degrees. Find Angle ABC.

****Q4.**** Show that angles in the same segment of a circle are equal. Use this to find an unknown angle given others.

****Q5.**** AB is a diameter of a circle, and AC is a chord. If Angle CAB = 30 degrees, prove that the radius of the circle is equal to half the length of chord BC.

****Q6.**** Two chords AB and CD intersect inside the circle. Using the property of angles in a segment, find a relation between two given angle measures.

****CHAPTER 17: TANGENTS AND SECANTS****

****YouTube Lecture:**** https://youtu.be/6_iEGczwfO8?si=JEFgL9UENGR8WJiu

****1. Tangent to a Circle****

A tangent is a line that touches the circle at exactly one point. The point of contact is the point where the tangent meets the circle.

****2. Key Theorems on Tangents and Secants****

****Radius Tangent Perpendicularity:**** The tangent at any point of a circle is perpendicular to the radius through the point of contact.

****Tangents from an External Point:**** The lengths of two tangents drawn from an external point to a circle are equal.

$$PA = PB$$

****Alternate Segment Theorem:**** The angle between a tangent and a chord through the point of contact is equal to the angle in the alternate segment.

****3. Secant Properties****

****Intersecting Chords Theorem (Internal):**** If two chords AB and CD intersect at point P inside the circle, then:

$$AP \times PB = CP \times PD$$

****Intersecting Secants Theorem (External):**** If two secants PAB and PCD are drawn from an external point P, then:

$$PA \times PB = PC \times PD$$

****Tangent Secant Theorem:**** If a tangent PT and a secant PAB are drawn from an external point P, then:

$$PT^2 = PA \times PB$$

****IMPORTANT QUESTIONS: TANGENTS AND SECANTS****

****Q1.**** Prove that the tangent at any point of a circle is perpendicular to the radius through the point of contact.

****Q2.**** From an external point P, two tangents PA and PB are drawn to a circle with centre O. Prove that Angle AOB and Angle APB are supplementary.

****Q3.**** Using the Alternate Segment Theorem, find the value of an angle between a chord and tangent given some other angle in the figure.

****Q4.**** In the given diagram, PT is a tangent and PAB is a secant to a circle. If PT = 6 cm, PA = 4 cm, find PB.

****Q5.**** Two chords AB and CD of a circle intersect at P inside the circle. If AP = 4 cm, PB = 6 cm, CP = 3 cm, find PD.

****Q6.**** Prove that the lengths of two tangents drawn from an external point to a circle are equal.

****Q7.**** Two circles touch externally at point C. Prove that the common tangent at C bisects the common internal tangent at points A and B where it touches the two circles.

****CHAPTER 18: CYLINDER, CONE AND SPHERE (MENSURATION)****

****YouTube Lecture:**** <https://youtu.be/GlcnVhwJA5Y?si=Z7zgiBza32MRuOuw>

****1. Cylinder (Right Circular Cylinder)****

Curved Surface Area (CSA) = $2 \times \pi \times r \times h$

Total Surface Area (TSA) = $2 \times \pi \times r \times (r + h)$

Volume = $\pi \times r^2 \times h$

****2. Cone (Right Circular Cone)****

Slant Height (l): $l = \sqrt{r^2 + h^2}$

Curved Surface Area (CSA) = $\pi \times r \times l$

Total Surface Area (TSA) = $\pi \times r \times (l + r)$

Volume = $(1/3) \times \pi \times r^2 \times h$

****3. Sphere and Hemisphere****

****Sphere:****

$$\text{Surface Area} = 4 \times \pi \times r^2$$

$$\text{Volume} = \left(\frac{4}{3}\right) \times \pi \times r^3$$

****Hemisphere (Solid):****

$$\text{Curved Surface Area} = 2 \times \pi \times r^2$$

$$\text{Total Surface Area} = 3 \times \pi \times r^2$$

$$\text{Volume} = \left(\frac{2}{3}\right) \times \pi \times r^3$$

****4. Combination of Solids and Melting/Recasting****

When one solid is melted and recast into another shape, the volume of the new solid equals the volume of the original solid. The surface areas will change.

****IMPORTANT QUESTIONS: CYLINDER, CONE AND SPHERE (MENSURATION)****

****Q1.**** Find the volume, curved surface area and total surface area of a cylinder with radius 7 cm and height 20 cm.

****Q2.**** The curved surface area of a cone is 550 cm^2 . If its slant height is 25 cm, find its radius and height. Then find its volume.

****Q3.**** A sphere of radius 6 cm is melted and recast into a cone of height 8 cm. Find the radius of the cone.

****Q4.**** A solid is composed of a cylinder with hemispherical ends. The total height of the solid is 20 cm, and the diameter of the cylinder and hemispheres is 14 cm. Find the total surface area and volume of the solid.

****Q5.**** A cylinder is open at both ends and is made of 1.5 cm thick metal. Its external diameter is 12 cm and its height is 90 cm. Find the volume of metal used to make the cylinder.

****Q6.**** A cone and a sphere have equal volumes and equal radii. Find the ratio of the height of the cone to its radius.

****CHAPTER 19: TRIGONOMETRIC IDENTITIES****

****YouTube Lecture:**** https://youtu.be/sKVWQK2Zn34?si=Kc4aMnbeAi_o21n0

****1. Basic Trigonometric Ratios****

$$\sin A = \frac{\text{Perpendicular}}{\text{Hypotenuse}}$$

$$\cos A = \frac{\text{Base}}{\text{Hypotenuse}}$$

$\tan A = \text{Perpendicular} / \text{Base}$
 $\operatorname{cosec} A = 1 / \sin A$
 $\sec A = 1 / \cos A$
 $\cot A = 1 / \tan A$

****2. Reciprocal Relations****

$\tan A = \sin A / \cos A$
 $\cot A = \cos A / \sin A$

****3. Fundamental Trigonometric Identities****

****Identity 1:**** $\sin^2 A + \cos^2 A = 1$
****Identity 2:**** $1 + \tan^2 A = \sec^2 A$
****Identity 3:**** $1 + \cot^2 A = \operatorname{cosec}^2 A$

****4. Proving Algebraic Trigonometric Expressions****

Use the fundamental identities to simplify and prove equations. General approach: convert all terms to $\sin A$ and $\cos A$, then simplify.

****IMPORTANT QUESTIONS: TRIGONOMETRIC IDENTITIES****

****Q1.**** Prove that: $(\sin A + \cos A)^2 + (\sin A - \cos A)^2 = 2$

****Q2.**** Prove the identity: $(1 + \tan^2 A) / (1 + \cot^2 A) = \tan^2 A$

****Q3.**** Prove that: $(\operatorname{cosec} A - \sin A)(\sec A - \cos A) = 1 / (\tan A + \cot A)$

****Q4.**** Prove: $\sqrt{(1 + \sin A) / (1 - \sin A)} = \sec A + \tan A$

****Q5.**** Evaluate: $(\sin 30^\circ + \cos 60^\circ) / (\sec 45^\circ + \tan 45^\circ)$

****Q6.**** Prove that $\sin^4 A + \cos^4 A = 1 - 2 \sin^2 A \cos^2 A$

****Q7.**** If $\sec A = 17/8$, find the values of all other trigonometric ratios.
